

50 Practice Questions

SE Crash Course · Lectures 3–6

These are the same shape as the real quiz: one recall question about the process map or the BTC case, and the rest asking you to judge something the way you did in the exercises. Each question has four options and a worked explanation.

Topic	Questions
Recall	1, 22, 23, 25, 26, 44, 49
Evidence (E / N / X)	2, 3, 8, 27, 28, 43
Boundary (I / E / A / naming)	4, 5, 6, 29, 30, 45
Stakeholders	7, 41, 42
Requirements (D / V / C / OK)	9–14, 31–34, 46, 48
Verification (structure + methods)	15–19, 24, 35–38, 47, 50
Traceability matrix	20, 21, 39, 40

Set 1 — Questions 1 to 25

Recall · Evidence · Boundary ·
Requirements · Verification ·
Traceability

1. Recall

Which of these is a Stage 1 artifact?

- A. Stakeholder requirement
- B. Technical requirement on the Physical I/O Diagram
- C. Context diagram**
- D. Verification table

Correct answer: C — Stage 1 is problem definition. Its artifacts are user needs, evidence, and the context diagram. Stakeholder requirements come in Stage 2. Technical requirements with numbered components come in Stage 3.

2. Evidence

"We lose about two hours a week working out what shipped against what actually turned up." — receiving plant coordinator. What is it?

- A. A need
- B. A solution
- C. A stakeholder requirement
- D. Evidence**

Correct answer: D — Attributed, verbatim, written down as spoken — that is the definition of evidence. Turning it into a need is the next step: 'The receiving plant must be able to confirm what arrived without manual reconciliation.'

3. Evidence

"The system needs an RFID reader mounted at the trailer door." What is this?

- A. Evidence
- B. A need
- C. An X — it has already chosen a solution**
- D. A stakeholder requirement

Correct answer: C — It names a technology (RFID reader) and a location (trailer door). The need underneath is that the loader must be able to record what went aboard without stopping. The RFID reader is one way to achieve that — not a need.

4. Boundary

In the As-Is context diagram, which entity is classified as an Answer (A) rather than External (E)?

- A. Radio regulator
- B. Trailer
- C. BTC vehicle tracking system
- D. Barcode scanner**

Correct answer: D — The barcode scanner is an answer — it only exists if you choose to build a solution that uses one. It is already ruled out by the problem. The radio regulator, trailer, and BTC vehicle tracking system all exist whether or not you build anything.

5. Boundary

Which entity is added to the To-Be context diagram that does not appear in the As-Is?

- A. Loader
- B. Trailer
- C. Logistics customer
- D. Cargo location record**

Correct answer: **D** — Cargo location record (SYS.1) is the one box added — it is the system of interest. Every external entity in the As-Is diagram is unchanged. One box was added. That is the contribution.

6. Boundary

What is the correct way to name the system box in a To-Be context diagram?

- A. By the main technology it uses, e.g. "RFID Tracking System"
- B. By what it achieves, in three words or fewer, with no technology name**
- C. By the company that commissioned it
- D. By the primary user role it serves

Correct answer: **B** — "RFID Tracking System" fails because it names a technology. The box should say what the system achieves — "Cargo location record" or "Cargo visibility service" work. Naming a technology locks in a design before the problem is understood.

7. Stakeholders

Which stakeholder category covers people who are legally responsible if something goes wrong?

- A. Owners and sponsors
- B. Maintainers
- C. Safety and compliance**
- D. Affected non-users

Correct answer: **C** — Safety and compliance stakeholders are people legally responsible if something goes wrong — the radio regulator in the BTC case. Owners and sponsors pay for the system. Maintainers repair and calibrate it. Affected non-users are constrained by it.

8. Stakeholders

What is an "orphan need"?

- A. A need that cannot be turned into a stakeholder requirement
- B. A need with no evidence source — nobody has said it yet**
- C. A need that has been rejected by the client
- D. A need covered by two conflicting requirements

Correct answer: **B** — An orphan need has no source extract to link it to. The right response is not to delete it — it is to go and ask someone. It is a question nobody has asked yet.

9. Requirements

"Device allows an RFID module to communicate with a GPS tracking system." Which criterion does it fail?

- A. Verifiable
- B. Design independent**
- C. Correct
- D. It fails none

Correct answer: **B** — It names two technologies (RFID module, GPS tracking system) and an architecture. This is a design written as a requirement. Design independence asks: could someone satisfy this more than one way? No — there is only one prescribed way.

10. Requirements

"Device can read RFID tags at a range of up to 5 metres." Which criterion does it fail?

- A. Design independent
- B. Correct
- C. Verifiable**
- D. Complete

Correct answer: **C** — 'Up to' points the wrong way. A reader that manages one centimetre satisfies this requirement. No threshold can distinguish a passing device from a failing one. It should say 'at least 5 metres'.

11. Requirements

"Device will queue product movements when an unusable (absent or low) GSM signal is present." Which criterion does it fail?

- A. Design independent
- B. Correct
- C. Verifiable**
- D. Complete

Correct answer: **C** — Names GSM, and 'unusable', 'low', and 'reliable' have no thresholds. Right instinct — the queueing behaviour is needed — but two engineers would start queueing at different signal levels. Unverifiable as written.

12. Requirements

"Requires little to no vehicle-operator input for operation." Which criterion does it fail?

- A. Correct
- B. Design independent
- C. Verifiable**
- D. Complete

Correct answer: **C** — 'Little to no' cannot be tested. Two people would not agree on whether it passed. A verifiable requirement needs a threshold that anyone could measure the same way.

13. Requirements

"Device is capable of storing at least 1000 tags of data." How does this requirement score?

- A. Fails verifiable — no test method is specified
- B. Fails design independent — it names a technology
- C. Fails correct — the arithmetic is wrong
- D. Passes all four criteria**

Correct answer: **D** — Measurable, has a threshold ('at least 1000'), names no technology, and contains no false arithmetic. This is the best of the ten BTC requirements. It is the only one marked OK.

14. Requirements

"Utilizes the vehicle's power to operate." Which criterion does this fail?

- A. Verifiable
- B. Design independent**
- C. Correct
- D. Complete

Correct answer: **B** — A design choice (take power from the vehicle) is presented as a requirement. It also contradicts the need for the record to survive while the trailer sits alone and detached. Design independence asks whether someone could satisfy this need in more than one way — yes, a battery would do.

15. Verification

A properly formed verification must contain which three things?

- A. A method, a pass criterion, and a named engineer
- B. Evidence that will exist afterwards, a method, and a pass criterion written before anyone runs it**
- C. A requirement reference, a test date, and a signature
- D. An instrument, a technician, and a numeric threshold

Correct answer: **B** — Evidence (something that exists afterwards that anyone can look at), Method (a named approach), and Pass Criterion (decided before the test, not after). The criterion written after the test is worthless — you already know the answer.

16. Verification

"For the purposes of this project, this requirement is fulfilled by an RFID reader from Alien Technologies." What is this?

- A. A valid inspection-method verification
- B. Evidence with no method or criterion
- C. Not a verification — it names the part they bought**
- D. A demonstration-method verification

Correct answer: **C** — Nothing was done, nothing was measured, and nothing exists afterwards that anyone can look at. A decision was made, but decisions are not evidence. An inspection of the reader's compliance certificate would have satisfied all three parts in ten minutes.

17. Verification

Which verification method involves operating the real system with no instruments, just watching it work?

- A. Test
- B. Analysis
- C. Demonstration**
- D. Modelling

Correct answer: **C** — Demonstration = watch it work, informally, no instruments. Loading a trailer and seeing the manifest change is a demonstration. A test adds instruments and records numbers. Analysis calculates it. Modelling simulates the real thing.

18. Verification

"The system shall retain the record of trailer contents for at least 72 hours while separated from the vehicle." The verification: "Disconnect the trailer unit, wait 72 hours, read the record. Evidence: the record, with timestamps. Pass: contents unchanged and complete." Which method?

- A. Inspection
- B. Analysis
- C. Modelling
- D. Test**

Correct answer: **D** — It runs the system under controlled conditions and records measurable results (timestamps, contents). That is a test. It measures — with evidence you can check afterwards.

19. Verification

"The HCP65-G has 1.7 MB of space to store log files in, and the microcontroller is interfaced with a 4 Mbit flash chip." Cited against: "The system will have at least 3.02 KB of memory specifically for data packets." What is the problem?

- A. The verification uses the wrong method
- B. There is no pass criterion
- C. The requirement itself is wrong — 33 bytes x 1000 tags = 33 KB, not 3.02 KB**
- D. The verification does not name a method

Correct answer: **C** — V5 is a perfect verification of a defective requirement. The method (analysis) is sound. Evidence is the datasheet figures. The criterion is 3.02 KB from the requirement. But 33 bytes x 1000 tags = 33 KB — the requirement is wrong by a factor of ten.

20. Traceability

In the BTC traceability matrix, requirements 7 and 8 have no ticks at all. What should this tell you?

- A. Those requirements are optional
- B. Those requirements were verified informally and not recorded
- C. No verification addresses them — they were never verified**
- D. Those requirements contradict each other

Correct answer: **C** — Requirements 7 (queueing when signal is unusable) and 8 (no operator input) have no verifications. Nobody noticed because nobody built this matrix until the end. A traceability matrix finds things reading requirements line by line does not.

21. Traceability

In the matrix, V3 is ticked against requirements 2, 3, and 4. What should you suspect?

- A. V3 is an especially thorough test
- B. Requirements 2, 3, and 4 are duplicates
- C. V3 probably addresses one of them and is being borrowed for the other two**
- D. Nothing — this is normal and healthy

Correct answer: **C** — When one verification covers multiple requirements it is usually doing the real work for one and being cited for the rest. In the BTC case V3 only addresses storage — it was borrowed for transmission and direction, which it does not actually test.

22. Recall

Which correctly describes the difference between an As-Is and a To-Be context diagram?

- A. As-Is shows stakeholders; To-Be shows requirements
- B. As-Is shows the world without the system; To-Be adds one box for the system of interest**
- C. As-Is is drawn by the client; To-Be is drawn by the engineer
- D. As-Is is optional; To-Be is mandatory

Correct answer: B — The As-Is shows the problem as it exists today — the people and flows with no system yet. The To-Be adds exactly one box: the system you are about to build. Every external entity remains unchanged. The contribution is one box and what crosses its boundary.

23. Recall

What disappeared from the BTC As-Is diagram when the To-Be was drawn?

- A. The receiving plant
- B. The discrepancy report
- C. The logistics customer
- D. The handwritten load note**

Correct answer: D — The handwritten load note disappears. A person stops having to write something down. That — not the technology — is the contribution. The discrepancy report also disappears, but the handwritten load note is the clearest answer to what the loader no longer has to do.

24. Verification

Which is a correctly formed verification for: "The system shall identify an item at a distance of at least 5.0 metres from the reading point"?

- A. The reader is a compliant model from a reputable supplier.
- B. We will test the reader thoroughly before installation.
- C. Place a tagged item at 5.0 m and read it 100 times. Evidence: count of successful reads. Pass: 100 of 100.**
- D. The reader works properly.

Correct answer: C — It has all three parts: a method (place and read), evidence that will exist afterwards (count of successful reads), and a pass criterion written before the test (100 of 100). The other options name a part, promise future action, or make a vague assertion.

25. Recall

"BTC must be able to build further units without the original design team." What is this?

- A. Evidence
- B. A solution
- C. A need**
- D. A stakeholder requirement

Correct answer: C — This is an owner's need. Not everything comes from users — owners have needs too. It says what someone must be able to do, with no solution in it. Needs do not all come from operators at the sharp end.

Set 2 — Questions 26 to 50

No repeats from Set 1 —
different angles on the same
topics

26. Recall

What does Stage 3 of the SE process add that Stage 2 does not?

- A. User needs and evidence
- B. Stakeholder requirements
- C. Numbers attached to components on the Physical I/O Diagram**
- D. A verification table

Correct answer: C — Stage 3 is where subsystems become components and components get numbers attached. That is what makes the Physical I/O Diagram different from earlier stages — it names and numbers every part.

27. Evidence

Which of these is correctly classified as a Need?

- A. "If I have to stop and scan every pallet I am doing half the loads." — forklift operator
- B. The system needs an RFID reader mounted at the trailer door.
- C. The loader must be able to complete a load without stopping to record what went on.**
- D. The system shall record each load event within two seconds.

Correct answer: C — A need says what someone must be able to do with no solution in it. Option A is evidence (verbatim, attributed). Option B names a technology — that is an X. Option D is a requirement, not a need — right idea, wrong document, and premature.

28. Evidence

"The system shall record each load event within two seconds." What is this?

- A. Evidence
- B. A need
- C. An X — it is a requirement, not a need, and it is premature**
- D. A correctly formed stakeholder requirement

Correct answer: C — Right idea — there is a real need for timely recording. But this is a requirement, not a need. It belongs in a requirements document, not a needs list. Writing it here skips the step of understanding what the need actually is.

29. Boundary

How is the RFID reader classified in the As-Is context diagram?

- A. Internal (I) — part of what you are designing
- B. External (E) — it interacts but you cannot change it
- C. Answer (A) — a component of one possible solution, not yet part of the problem**
- D. It should not appear on the diagram at all

Correct answer: C — The RFID reader is an Answer — it only exists if you choose to build a solution that includes one. The As-Is shows the problem as it is today. Putting an RFID reader in the As-Is means you have already decided how to solve the problem.

30. Boundary

Which of these is an acceptable name for the system box in the To-Be diagram?

- A. RFID Tracking System
- B. Trailer Scanner
- C. Cargo Visibility Service**
- D. GPS Asset Monitor

Correct answer: **C** — *Cargo Visibility Service* says what it achieves — it is silent on how. *RFID Tracking System* names a technology. *Trailer Scanner* names a mechanism (scanning). *GPS Asset Monitor* names another technology. Purpose, not mechanism.

31. Requirements

"Has a backup power source." Which criterion does this fail?

- A. Verifiable
- B. Design independent**
- C. Correct
- D. Complete

Correct answer: **B** — *It names the answer. The underlying need is that the record survives while the trailer sits alone. A backup power source is one way to achieve that — a capacitor or a long-life primary cell would be others. The requirement has already chosen the mechanism.*

32. Requirements

What does the "Correct" criterion specifically ask you to check?

- A. Whether two engineers would measure the same result
- B. Whether the requirement could be satisfied more than one way
- C. Whether any arithmetic in the requirement holds**
- D. Whether the requirement covers every failure scenario

Correct answer: **C** — *Correct asks: does any arithmetic in it hold? Option A describes verifiable. Option B describes design independent. Option D describes complete. The V5 requirement in the BTC case failed correct — 33 bytes x 1000 tags = 33 KB, not 3.02 KB.*

33. Requirements

"Device stores RFID tag data with GPS location and time data in memory for transmission to a web server." Which criterion does it fail?

- A. Correct
- B. Verifiable
- C. Design independent**
- D. Complete

Correct answer: **C** — *Three technologies named in one sentence: RFID tag, GPS, web server. The need underneath is simply that a record of what is aboard reaches BTC's servers. How it gets there is a design question.*

34. Requirements

The four criteria for a well-formed stakeholder requirement are: design independent, verifiable, correct, and ____?

- A. Complete
- B. Traceable
- C. Unambiguous
- D. Testable

Correct answer: **A** — The four are D (design independent), V (verifiable), C (correct), and OK — which stands for nothing wrong, i.e. complete. Complete asks: what could go wrong that this requirement does not cover?

35. Verification

Which verification method do you use when the real system is too expensive, dangerous, or not yet built?

- A. Test
- B. Inspection
- C. Demonstration
- D. Modelling

Correct answer: **D** — Modelling = simulate it. It is for situations where running the real system is impractical — signal coverage across a route before any hardware exists is a BTC-case example. The model stands in for the physical thing.

36. Verification

"We will test the reader thoroughly before installation." What is wrong with this as a verification?

- A. It uses the wrong method — inspection would be better
- B. It promises future action but contains no method, evidence, or pass criterion
- C. It names a part instead of describing a test
- D. Nothing — future-tense verifications are acceptable

Correct answer: **B** — A statement of intent is not a verification. There is no method (how?), no evidence (what exists afterwards?), and no pass criterion (what would a pass look like?). It could mean anything from a two-second power-on check to a hundred-run field trial.

37. Verification

"The HCP65-G has a GPS receiver, and we have verified that it works properly (see Section 4.1.1 for test results)." Which part of verification is missing?

- A. Evidence
- B. Method
- C. Pass criterion
- D. All three parts

Correct answer: **C** — Nobody said beforehand what 'works properly' would mean. Nobody could look at the result and say whether it was a pass, because nobody stated what a pass would be. Evidence exists (test results). A method was implied. But the criterion was never written.

38. Verification

Which verification method involves checking configuration, dimensions, labels, part numbers, or documents — without running the system?

- A. Test
- B. Analysis
- C. Demonstration
- D. Inspection**

Correct answer: D — Inspection = look at it, without running it. Checking that the RFID reader's certificate says ISO 18000-6C is an inspection. It is usually the cheapest method when documentation or physical configuration can answer the question.

39. Traceability

Requirements 7 and 8 in the BTC matrix had no ticks. Nobody noticed. What is the root cause?

- A. Requirements 7 and 8 were impossible to verify with available equipment
- B. The matrix was built at the end, not throughout — nobody was looking at the whole picture**
- C. The verifications were filed in a separate appendix
- D. Requirements 7 and 8 were added after the verifications were written

Correct answer: B — Nobody built this matrix until the end, and then nobody read it carefully. A traceability matrix finds things that reading requirements line by line does not — empty rows in ten minutes. Building it once, by hand, is what makes the generated one readable later.

40. Traceability

What does a traceability matrix reveal that reading the requirements document line by line does not?

- A. Whether each requirement is correctly worded
- B. Which requirements have no verification, and which verifications are being borrowed across multiple rows**
- C. Whether stakeholder needs are fully captured
- D. The order in which verifications should be run

Correct answer: B — The matrix shows two things at once: rows with no ticks (orphan requirements — never verified) and columns cited against too many rows (borrowed verifications — doing the work of one and being claimed for the rest). Neither is visible from a linear read.

41. Stakeholders

Which of the eight stakeholder categories covers systems or teams that must interface with yours?

- A. Beneficiaries
- B. Lifecycle stakeholders
- C. Integrators**
- D. Affected non-users

Correct answer: C — Integrators are systems or teams that must interface with yours — the BTC vehicle tracking system is an integrator in the BTC case. Beneficiaries gain from the system but don't operate it. Lifecycle stakeholders interact in future phases. Affected non-users are constrained by it.

42. Stakeholders

Why should you name stakeholder roles rather than individual people?

- A. Privacy regulations prohibit naming individuals in engineering documents
- B. Roles survive staff changes; people do not**
- C. Individual people may have conflicting personal interests
- D. It keeps the context diagram from becoming cluttered

Correct answer: **B** — "Forklift operator", not "Bilal". If Bilal leaves, the role stays. Requirements, needs, and evidence tied to a role remain valid when the person in the role changes. Requirements tied to a named individual become orphaned the day they leave.

43. Evidence

Which is the correct sequence when building a requirements document from scratch?

- A. Requirements → Needs → Evidence
- B. Evidence → Needs → Requirements**
- C. Needs → Evidence → Requirements
- D. Evidence → Requirements → Needs

Correct answer: **B** — You gather evidence first (what stakeholders actually said), extract needs from it (what they must be able to do), then write requirements to satisfy those needs. Running the sequence backwards produces solutions before the problem is understood.

44. Recall

How many external entity boxes appear in the BTC As-Is context diagram?

- A. 5
- B. 6
- C. 7**
- D. 8

Correct answer: **C** — Seven: Loader (EXT.1), Trailer (EXT.2), Pallets and cargo (EXT.3), Driver (EXT.4), Receiving plant (EXT.5), BTC vehicle tracking system (EXT.6), Logistics customer (EXT.7). Seven entities, seven connections, and nobody in the middle — that is the point.

45. Boundary

In the BTC As-Is context diagram, what crosses the line from the Loader to the Driver?

- A. Vehicle position report
- B. Handwritten load note**
- C. Discrepancy report
- D. Arrival check

Correct answer: **B** — The loader writes down what went onto the trailer by hand and gives it to the driver. In the To-Be diagram, this flow disappears entirely — the cargo itself tells the system it is there. That, not the technology, is the contribution.

46. Requirements

"Device stores data from OEM RFID module for communication to the GPS unit." Which criterion does it fail?

- A. Verifiable
- B. Design independent**
- C. Correct
- D. Complete

Correct answer: **B** — Same problem as requirement 1, plus it names a supplier's module. Two technologies, one architecture, one supplier — all locked in before any need analysis is complete. The need is simply that a record reaches the server.

47. Verification

"Load a trailer normally while observing. Evidence: an observation sheet listing every action the loader and driver take. Pass: none of them was done for the system." Which method is this?

- A. Test
- B. Inspection
- C. Analysis
- D. Demonstration**

Correct answer: **D** — Demonstration = operate the real system, no instruments. You watch it work informally. No instruments measure anything — instead, an observer records what actions people take. The pass criterion is that neither loader nor driver did anything for the system.

48. Requirements

"Detects whether a product is entering or exiting the truck." Which criterion does it fail?

- A. Design independent — it implies a sensing mechanism
- B. Correct — a truck cannot be distinguished from a trailer
- C. Verifiable — there is no frequency or accuracy threshold stated**
- D. It fails none — this is the best-formed of the ten

Correct answer: **C** — Solution-neutral, which is good. But how often must it be right? There is no number, so nothing can fail this requirement. A device that detects correctly one time in a thousand satisfies it as written. It should say 'at least X% of the time'.

49. Recall

Why is the As-Is context diagram drawn before the To-Be?

- A. To identify which existing technologies to remove
- B. Because clients contractually require a baseline document
- C. To understand the existing flows and entities in the problem before committing to any solution**
- D. To show which external entities will become internal once the system is built

Correct answer: **C** — The As-Is forces you to see the problem as it is — who does what, what crosses what boundary, what information exists already. Only then can you say precisely what one new box changes. Skipping it means designing before understanding.

50. Verification

"This requirement is fulfilled by the HCP65-G. The data it receives is sent via GSM to a server hosted by BTC Solutions." What is the specific problem with this verification?

- A. It has no evidence
- B. It names a method but no pass criterion
- C. It describes the intended design, not what was actually built — the GSM link was never functional**
- D. It cites the wrong requirement

Correct answer: C — Section 4.2 of the same report states the link is not functional and data had to be loaded by hand. This verification was written from memory of the design intent, not from what was tested. It is not merely unverified — it is untrue.